

Support for the New Claims

The overall architecture recited in new independent claims 19 and 35 is summarized at page 8, lines 9-26, and is described in detail with reference to FIG. 1 at pages 13-15, in particular from page 14, line 2, through page 15. These passages describe the mate creation component 110, the transcoding components 120, the manifest files 121 pointing to unique interleaved combinations of chunks 113 of the ensemble 112, the ledger 122, the detection component 130 running a camera cuts detection algorithm 131 that devises time codes from the pirate webcasting and builds reconstructed manifest files 121', and the retrieval component 140 searching the ledger for manifest files equal to the reconstructed manifest files 121'. Original claims 1, 10, 11, 14 and 15 recite this architecture in claim form. In new claims 19 and 35, the manifest file found through the equality search is one of the recipient-associated timing-choice manifest files generated, delivered and recorded by the claimed process, includes at least one of the later mate camera-switch timings and matches the detected combination represented by the reconstructed manifest file. This complete chain from production to attribution is disclosed by original claim 1 read with page 12, line 16, through page 13, line 4, by pages 14-16, by original claims 11-15 and 17 and by Examples 3-5.

As to the production side, the structured list of edit instructions with per-cut out-point and in-point time codes and source-camera identifiers is described at page 16, lines 18-28, and at page 18, lines 13-17. Example 2 at pages 33-35, with Tables 1 and 2, sets out the source camera and the in-point and out-point time codes of each cut and describes extending the duration of Cut 2 by 10 frames so that the start of Cut 3 is delayed by 10 frames while synchronization is maintained from Cut 4 onwards. Example 2 thereby also supports new claims 20 and 21, in which the mate transition occurs at a later timing through extension of the selection of the first camera. Variation at a plurality of selected camera cuts and manifest files representing combinations of per-cut timing choices are supported at page 16, lines 1-17, where each further camera cut doubles the number of content versions and manifest files.

As to distribution and the record of associations, the chunk segmentation, content delivery network distribution with adaptive streaming and manifest tailoring recited in new claim 27 are described at page 8, lines 9-21, at page 14, lines 10-21, and in Method 200 at page 29, lines 19-27, where segmenting step 240 enables adaptive streaming across varying network conditions and device capabilities and generation step 250 produces manifest files tailored to end user devices and network conditions. Original claim 11 recites the same relationship in claim form. The mixing of reference and mate chunks with progressive assignment of individualized manifest files, recited in new claim 28, is described at page 18, line 23, through page 19, line 12, and at page 40, lines 5-7, and is recited in original claim 12. Unicast delivery, recited in new claim 33, is described at page 30, lines 18-20, and is recited in original claim 6. The ledger identifying which end user received

which manifest file appears at page 8, lines 15-21, and at page 14, lines 17-21. See also original claims 13 and 14.

As to the detection side, including monitor-side method claim 38, the operations of receiving the suspected unauthorized distribution, applying a scene-change detection algorithm to identify time codes of camera cuts, building one or more reconstructed manifest files from the identified time codes and searching the ledger to identify the associated recipient are described at page 8, lines 15-21, at page 14, line 22, through page 15, and in method form at page 30, lines 1-10, describing the ledger recording step 270, the monitoring step 280 detecting scene changes and devising time codes and the searching step 290. Original claims 15, 16 and 17 recite the same operations in claim form.

As to the remaining dependent claims, the perceptual hash and fuzzy matching comparisons of new claims 25, 26, 36 and 37 are described at page 30, lines 21-28, in Example 5 at pages 40-42 and in original claims 2 and 3. Probabilistic Tardos fingerprinting for collusion scenarios, new claims 29 and 30, is described at pages 24-26 and in original claim 5. New claim 34 recites overlaying additional audio or video elements before segmentation into the chunks. The overlay is described at page 24, lines 12-16, and at page 30, lines 11-13, which place the overlay in programming step 230, and Method 200 at page 29, lines 15-24, places programming step 230 before segmenting step 240, thereby disclosing the claimed order. Original claim 4 separately confirms the overlay feature. The Edit Decision List implementation of the structured list, new claim 22, is described at page 16, lines 18-25. Capture of live video from diverse viewpoints, new claim 23, is recited in original claim 16. The computing environment recited in the preambles is described at page 18, lines 1-3, and in original claim 8.

New claim 24 is supported by page 16, lines 1-8, describing further camera cuts spawning additional mates and manifest files, read together with Example 2 at pages 33-35, which provides the concrete single-cut variation at the Cut 2 and Cut 3 boundary. New claim 31 is supported by page 15, lines 12-16, which states expressly that the reference and mate manifest files point to sets of chunks of equal duration, together with Example 3 at pages 36-38, which shows the respective manifest files selecting different chunks at corresponding positions. New claim 32 is supported by the corresponding reference and mate chunk positions of Example 3 read together with the source camera sequence and the different reference and mate switch timings of Example 2. For claims 24 and 32 the recited relationships are shown by the cited passages read together rather than by a single passage.